

CubeParam

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What is this?

CubeParam is a project that lets the user define and view a 3D object from any conceivable angle. The angle is controlled by a complex parameter $\alpha \in \mathbb{C}$ which determines the projection in $\mathbb{R}^2$ of points $(a, b, c) \in \mathbb{R}^3$. The user controls α by moving a pointer on screen.

Viewing Options

There are four basic viewing options: the user can toggle between two modes *Compact/Polynomial* and two anchors *z-Axis/Diagonal*. The projections maps are all linear: the projections $x_\alpha, y_\alpha, z_\alpha \in \mathbb{R}^2$ of the three axes determine the projection of any point:

$$(a, b, c) \mapsto ax_\alpha + by_\alpha + cz_\alpha \in \mathbb{R}^2$$

In Polynomial mode, the pointer is α . Here $x_\alpha, y_\alpha, z_\alpha$ are polynomials in α (here ζ denotes $\zeta_3 = e^{2\pi i/3} = \frac{-1+\sqrt{3}i}{2}$):

$$\begin{array}{cc} \text{Polynomial + z-Axis} & \text{Polynomial + Diagonal} \\ \left[\begin{array}{c} x_\alpha \\ y_\alpha \\ z_\alpha \end{array} \right] = & \frac{1}{2} \left[\begin{array}{c} 1 - \alpha^2 \\ i(1 + \alpha^2) \\ 2\alpha \end{array} \right] & -\frac{i}{3\sqrt{2}} \left[\begin{array}{c} 1 + \sqrt{2}i\alpha + \alpha^2 \\ \zeta^2 + \sqrt{2}i\alpha + \zeta\alpha^2 \\ \zeta + \sqrt{2}i\alpha + \zeta^2\alpha^2 \end{array} \right] \end{array} \quad (1)$$

Compact mode differs from Polynomial mode in two ways: Firstly, the pointer lives in the open unit disk. The pointer - let's call it α' - is related to α via

$$\alpha' = \frac{2}{\pi} \arctan |\alpha| \cdot \frac{\alpha}{|\alpha|}.$$

Secondly, $x_\alpha, y_\alpha, z_\alpha$ in Polynomial mode are replaced with $\frac{x_\alpha}{r_\alpha}, \frac{y_\alpha}{r_\alpha}, \frac{z_\alpha}{r_\alpha}$, where

$$r_\alpha = \begin{array}{cc} \text{Polynomial + z-Axis} & \text{Polynomial + Diagonal} \\ \frac{1+|\alpha|^2}{2} & \frac{1+|\alpha|^2}{3\sqrt{2}} \end{array} \quad (2)$$

These normalized vectors are projections of an *orthonormal* basis in $\mathbb{R}^3$. Perspective is also available to aid with depth perception. This makes use of the

depth coordinate $h_{x/y/z,\alpha}$ for each axis:

$$\begin{bmatrix} h_{x,\alpha} \\ h_{y,\alpha} \\ h_{z,\alpha} \end{bmatrix} = \begin{array}{c} \text{Polynomial + z-Axis} \\ \begin{bmatrix} -\Re(\alpha) \\ -\Im(\alpha) \\ \frac{1-|\alpha|^2}{2} \end{bmatrix} \end{array} \quad \frac{1}{6} \begin{array}{c} \text{Polynomial + Diagonal} \\ \begin{bmatrix} 1 - |\alpha|^2 + 2\sqrt{2} \Im(\alpha) \\ 1 - |\alpha|^2 + 2\sqrt{2} \Im(\zeta\alpha) \\ 1 - |\alpha|^2 + 2\sqrt{2} \Im(\zeta^2\alpha) \end{bmatrix} \end{array} \quad (3)$$

When perspective is enabled, the user controls the scaling factor P . In Compact mode, the focal distance is $F = 1/P$. In Polynomial mode, the focal distance scales with r_α : $F = r_\alpha/P$. Either way (taking $r_\alpha := 1$ in Compact mode) the point $r_\alpha(u, h) = (r_\alpha u, r_\alpha h) \in \mathbb{C} \times \mathbb{R}$ is projected onto $r_\alpha \frac{1}{1-Ph} u \in \mathbb{C}$.

What to Display

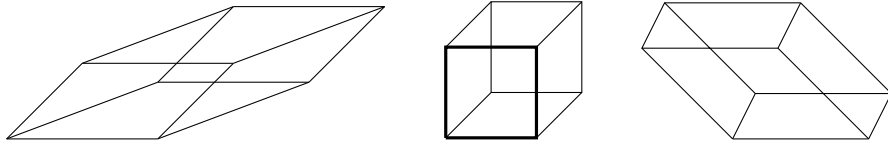
The user chooses what objects to display and how to display it:

- Define points, segments, polygonal faces, parametric curves.
- Numerical input supports symbolic expressions.
- Use auxiliary variables for versatile control of numerical, color, and boolean values.

Motivation

Classifying Drawings of Cubes

The motivation for this project is a simple question: what does a cube look like? Orthographic projection is a fair approximation of light hitting the retina. The problem of characterizing orthographic projections of cubes is considerably simpler. Consider these three parallelepipeds in $\mathbb{R}^2$:



Which of these look like a cube? The one on the left looks too *stretched*, while the one on the right looks too *squished*. This we know from intuition, backed by experience. The one in middle is tricky, partly because this is how most of us were taught to draw cubes. But think about it: if one face of a cube, normally projected onto a parallelogram, looks like a perfect square, then the opposing face must be *directly behind it*. That's not the case here. So None of these is an orthographic projection of a cube. Then what *is*?

It takes three sides to define these parallelepipeds, so the question takes another form: Let $v_k = \begin{bmatrix} x_k \\ y_k \end{bmatrix} \in \mathbb{R}^2$ for $k = 1, 2, 3$. When are these v_k orthographic

projections of three perpendicular vectors of equal length in $\mathbb{R}^3$? This turns into a lovely little exercise in linear algebra, and the answer is: if and only if

$$x_1^2 + x_2^2 + x_3^2 = y_1^2 + y_2^2 + y_3^2 \quad \text{and} \quad x_1y_1 + x_2y_2 + x_3y_3 = 0.$$

A little finessing reveals that the v_k are best thought of as complex numbers: if $u_k = x_k + y_k i \in \mathbb{C}$, then the parallelepiped spanned by the v_k is the orthographic of a cube if and only if

$$u_1^2 + u_2^2 + u_3^2 = 0. \tag{4}$$

Projective Algebraic Geometry over $\mathbb{C}$

Observation: if $(u_1, u_2, u_3) \in \mathbb{C}^3$ is a solution to (4), and if $\lambda \in \mathbb{C}$, then $(\lambda u_1, \lambda u_2, \lambda u_3)$ is also a solution to (4). This is no coincidence for two reasons: algebraically, λ^2 pops out of the left hand side; geometrically, multiplying by a non-zero complex number is simply a combined rotation and rescaling - so what was a cube drawing will *remain* a cube drawing. In mathematical jargon, the set of cube drawings - up to rotation and rescaling - correspond to the points on the complex projective curve X defined by (4):

$$X = \{(u_1, u_2, u_3) \in \mathbb{C}^3 - \{0\} \mid u_1^2 + u_2^2 + u_3^2 = 0\} / \mathbb{C}^\times \subseteq \mathbb{P}_{\mathbb{C}}^2.$$

This is a well-understood geometric object: X is a smooth conic projective curve, isomorphic to $\mathbb{P}_{\mathbb{C}}^1$, the Riemann sphere $\mathbb{C} \cup \{\infty\}$. An isomorphism from $\mathbb{P}_{\mathbb{C}}^1$ to X can be given as a triple of polynomials

$$\mathbb{P}_{\mathbb{C}}^1 \xrightarrow{\sim} X \quad \alpha \mapsto (f_x(\alpha), f_y(\alpha), f_z(\alpha))$$

of degree ≤ 2 . For example, the formula in Polynomial + z-Axis is obtained from the well-known classification of Pythagorean triples by manipulating (4) into

$$u_1^2 + u_3^2 = -u_2^2 = (iu_2)^2.$$

The formula in Polynomial + Diagonal takes into account the symmetry of X that comes from permuting the u_k .

What's Next?

In CubeParam

Thoughts for future implementation in CubeParam:

- Design and implement version for small screen (phone).
- Face drawing requires fragmenting the faces (BSP algorithm). When rendered with WebKit, this leaves visible seams between face fragments. This should be fixed. Rendering with Blink is seamless. Pun intended.

- Proper cohabitation on canvas of faces with vertices and segments.
- Surfaces, either parametric or as solutions to implicit equations
- Introduction of light source that is fixed by tablet compass.

Cubes on Grids

This project would use the underlying math of CubeParam to classify all orthographic projections of cubes with vertices on a square or triangular lattice. We tweak the isomorphisms $\mathbb{P}^1 \xrightarrow{\sim} X$ defining both Polynomial modes and get

$$\varphi_i: \mathbb{P}^1 \xrightarrow{\sim} X, \quad \alpha \mapsto (1 - \alpha^2, i(1 + \alpha^2), 2\alpha),$$

$$\varphi_\zeta: \mathbb{P}^1 \xrightarrow{\sim} X, \quad \alpha \mapsto (1 + 2\alpha - 2\alpha^2, \zeta^2 + 2\alpha - 2\zeta\alpha^2, \zeta + 2\alpha - 2\zeta^2\alpha^2).$$

These are defined respectively over the fields $\mathbb{Q}(i)$ and $\mathbb{Q}(\zeta)$. After homogenizing these and restricting to integral points, we get

$$\Phi_i: \mathbb{Z}[i]^2 \rightarrow \mathbb{Z}[i]^3, \quad [s : t] \mapsto [t^2 - s^2 : i(t^2 + s^2) : 2st],$$

$$\Phi_\zeta: \mathbb{Z}[\zeta]^2 \rightarrow \mathbb{Z}[\zeta]^3, \quad [s : t] \mapsto [t^2 + 2st - 2s^2 : \zeta^2 t^2 + 2st - 2\zeta s^2 : \zeta t^2 + 2st - 2\zeta^2 s^2].$$

What we get with Φ_i and Φ_ζ are classifications of essentially all orthographic projections of cubes with vertices on the square lattice $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$ and the triangular lattice $\mathbb{Z}[\zeta] = \{a + b\zeta \mid a, b \in \mathbb{Z}\}$.

Visualizing Knot Colorings

This project would build on the display capabilities of CubeParam to visualize the knot-theoretical notion of *knot colorings*. Knot theory is a field of math that studies what might be described as tangles of closed loops of string. Knot coloring is a way to color the projection of a knot. The colorings of a knot K can tell us a lot about K . In this project we might input a 3D model of a knot K and a coloring c . As K rotates, the projected image will shift around, shifting the colors of c accordingly. The goal is to visualize this.